

SANA'A INTL, Yemen

WMO: 414040

Lat: 15.476N

Lon: 44.220E

Elev: 7237

StdP: 11.24

Time Zone: 3.00 (E03)

Period: 02-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 38.9	(c) 41.0	(d) -0.5	(e) 7.0	(f) 60.7	(g) 4.7	(h) 9.2	(i) 61.4	(j) 16.9	(k) 69.0	(l) 15.9	(m) 68.2	(n) 2.3	(o) 220	(p) 0.370

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 23.3	(c) 86.4	(d) 54.8	(e) 85.9	(f) 54.6	(g) 84.5	(h) 54.3	(i) 61.9	(j) 72.1	(k) 60.9	(l) 71.6	(m) 60.2	(n) 71.6	(o) 11.2	(p) 40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.0	98.1	64.9	57.5	92.6	65.1	56.6	89.8	65.2	31.6	72.2	30.9	71.9	30.2	71.6	66.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	16.0	14.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			25.6	64.0	3.2	1.3	23.3	65.0	21.5	65.8	19.7	66.5	17.4	67.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.2	58.9	62.2	65.3	67.4	70.4	73.3	74.1	72.4	69.7	62.9	60.1	57.5
	DBStd	6.21	2.96	3.59	3.15	2.33	1.92	1.87	2.16	2.11	2.62	2.39	2.82	3.32
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	774	189	88	34	7	0	0	0	0	1	73	148	234
	CDD50	5912	276	342	475	523	633	699	746	693	591	401	302	231
	CDD65	1211	0	10	44	80	168	249	281	228	142	9	0	0
	CDH74	12276	128	433	686	841	1608	2545	2405	1892	1411	203	83	41
CDH80	2640	1	33	65	84	277	767	759	453	201	0	0	0	
Wind	WSAvg	5.5	4.8	5.0	5.4	5.3	6.1	6.3	6.4	6.4	6.1	5.4	4.6	4.5
Precipitation	PrecAvg	7.4	0.0	0.1	1.0	1.6	0.7	0.3	1.3	2.2	0.2	0.2	0.1	0.0
	PrecMax	14.3	0.1	0.8	4.3	7.2	3.0	3.3	3.6	7.1	1.1	0.9	0.8	0.2
	PrecMin	2.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.3	0.0	0.2	1.2	1.5	0.8	0.7	0.9	1.7	0.2	0.2	0.2	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.0	82.8	84.1	84.1	85.9	88.1	88.2	86.4	85.8	79.1	78.3	77.1
		MCWB	50.3	52.5	53.2	53.7	53.9	55.3	55.6	56.2	55.1	50.8	51.2	51.3
	2%	DB	77.1	80.8	82.1	82.3	84.2	86.4	87.4	85.5	84.0	77.3	76.2	75.2
		MCWB	49.0	51.2	52.8	53.1	53.3	54.3	55.1	56.0	54.1	50.2	51.1	50.6
	5%	DB	75.4	79.0	80.3	80.7	82.7	85.8	85.9	84.2	82.3	76.6	74.9	73.5
		MCWB	49.2	50.5	52.2	53.0	53.2	53.9	54.7	55.8	53.6	50.0	51.0	49.4
	10%	DB	73.4	76.9	78.5	79.1	81.8	84.4	84.3	82.6	80.8	75.0	73.0	71.6
		MCWB	49.3	50.8	52.2	53.1	53.1	53.0	54.9	55.9	53.1	49.7	50.4	48.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.1	57.5	60.3	61.1	61.9	61.0	63.3	62.9	60.6	58.4	57.2	57.0
		MCDB	69.9	73.1	73.3	71.5	70.1	74.0	72.7	73.2	73.3	66.3	65.4	71.9
	2%	WB	54.0	55.8	58.7	59.9	60.3	59.5	61.8	61.9	59.3	56.3	55.1	55.5
		MCDB	68.9	71.3	71.5	70.7	70.8	75.3	72.4	72.0	73.0	65.9	67.7	69.1
	5%	WB	52.9	54.6	57.5	58.6	59.3	58.3	60.7	61.0	57.9	53.5	53.7	53.9
		MCDB	68.3	70.8	70.7	69.9	70.3	75.3	72.0	70.9	72.5	70.6	68.1	67.3
	10%	WB	51.5	53.2	56.2	57.6	58.1	57.2	59.8	60.2	56.6	51.2	52.3	51.9
		MCDB	67.8	70.3	69.7	69.3	70.1	74.8	72.4	70.9	72.7	72.7	68.8	66.8
Mean Daily Temperature Range	5% DB	MDBR	31.0	30.6	27.5	24.8	23.6	23.3	21.8	22.1	23.5	25.3	27.6	30.8
		MCDBR	32.7	31.9	29.4	26.7	24.8	24.5	23.4	23.3	23.9	26.3	28.9	31.5
		MCWBR	14.4	12.9	11.6	10.4	9.4	9.4	7.9	7.5	9.0	11.1	12.8	14.4
	5% WB	MCDBR	28.2	29.0	25.4	23.9	22.9	23.2	21.8	22.3	22.9	21.4	25.3	26.8
		MCWBR	13.0	11.8	9.6	8.9	8.3	9.7	7.6	7.2	9.2	10.3	11.5	12.3
Clear-Sky Solar Irradiance	taub	0.332	0.367	0.434	0.452	0.482	0.572	0.647	0.610	0.499	0.375	0.353	0.332	
	taud	2.383	2.253	2.046	2.013	1.913	1.677	1.581	1.655	1.859	2.207	2.287	2.366	
	Ebn at Noon	305	298	280	273	261	236	220	230	258	291	295	301	
	Edh at Noon	37	44	55	57	62	78	86	81	66	45	40	36	
All-Sky Solar Radiation	RadAvg	1794	1975	2120	2181	2253	2196	1956	1975	2064	2097	1873	1757	
	RadStd	57	79	104	121	98	95	66	64	69	73	56	69	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page